

Package ‘GR2MSemiDistr’

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Type Package

Title A package for hydrological modeling with a semi-distribute GR2M model adaptation in large-sample studies and operational systems.

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Description This package run a semi-distributed GR2M model adaptation using the Weighted Flow Accumulation algorithm in TauDEM_537 (required)

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Encoding UTF-8

Depends R (>= 4.0),

Imports airGR, foreach, hydroGOF, raster, terra, geos, rtop, tictoc, lubridate, sf, exactextractr, parallel,

LazyData true

RoxygenNote 7.3.2

Contents

Create_Forcing_Inputs	1
Optim_GR2MSemiDistr	3
Routing_GR2MSemiDistr	4
Run_GR2MSemiDistr	6

Index	8
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Create_Forcing_Inputs	<i>Extract and prepare model input data in the airGR format (DatesR, P, E, and optionally Q) using gridded monthly precipitation and evapotranspiration.</i>
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Description

This function extracts spatially averaged monthly precipitation and potential evapotranspiration values for each subbasin, and optionally includes observed streamflow data, to produce a formatted data frame compatible with the airGR hydrological modeling framework.

Usage

```
Create_Forcing_Inputs(
  Subbasins,
  Precip,
  PotEvap,
  Qobs = NULL,
  DateIni,
  DateEnd,
  IniGrids = "01/1981",
  Save = FALSE,
  Update = FALSE,
  Members = NULL
)
```

Arguments

Subbasins	A 'SpatVector' object containing the subbasins' geometries. Must include the following attributes: 'COMID' (unique identifier), and 'Region' (string code for the hydro-climatic region).
Precip	A 'SpatRaster' object with monthly precipitation data in [mm/month].
PotEvap	A 'SpatRaster' object with monthly potential evapotranspiration data in [mm/month].
Qobs	Optional. A vector with observed streamflow in [m ³ /s] at the basin outlet.
DateIni	Start date of the output time series in 'mm/yyyy' format.
DateEnd	End date of the output time series in 'mm/yyyy' format.
IniGrids	Initial date of the gridded precipitation and evapotranspiration datasets in 'mm/yyyy' format.
Save	Logical. If TRUE, saves the resulting data frame to a '.txt' file in the './Inputs' directory. Default is FALSE.
Update	Logical. If TRUE, returns only the last month of data (useful for operational updates). Default is FALSE.
Members	Optional. Integer indicating the number of ensemble members for streamflow forecasting. If provided, the date vector will be repeated accordingly. Default is NULL.

Value

A data frame containing the model inputs in the airGR format, with columns:

DatesR Monthly dates

P_x Mean precipitation for subbasin x, where x is the COMID

E_x Mean potential evapotranspiration for subbasin x, where x is the COMID

Q Observed streamflow [m³/s], if provided

References

Aybar, C., Fernández, C., Huerta, A., Lavado, W., Vega, F., & Felipe-Obando, O. (2020). Construction of a high-resolution gridded rainfall dataset for Peru from 1981 to the present day. *Hydrological Sciences Journal*, 65(5), 770–785. <https://doi.org/10.1080/02626667.2019.1649411>

Llauca, H., Lavado-Casimiro, W., Montesinos, C., Santini, W., & Rau, P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. <https://doi.org/10.3390/w13081048>

Examples

```
library(GR2MSemiDistr)
data(pisco_pr) # Monthly precipitation SpatRaster
data(pisco_pe) # Monthly potential evapotranspiration SpatRaster
data(qobs)      # Data frame with observed streamflow
data(roi)       # Subbasins SpatVector

# Create model input data
data_inputs <- Create_Forcing_Inputs(
  Subbasins = roi,
  Precip = pisco_pr,
  PotEvap = pisco_pe,
  Qobs = qobs,
  DateIni = '01/1981',
  DateEnd = '12/2016',
  IniGrids = '01/1981'
)
View(data_inputs)
```

Optim_GR2MSemiDistr	<i>Model parameter optimization with the SCE-UA algorithm.</i>
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Description

Model parameter optimization with the SCE-UA algorithm.

Usage

```
Optim_GR2MSemiDistr(
  Data,
  Subbasins,
  RunIni,
  RunEnd,
  WarmUp = NULL,
  Parameters,
  Parameters.Min,
  Parameters.Max,
  Max.Functions = 5000,
  Optimization = "NSE",
  No.Optim = NULL
)
```

Arguments

Data	Dataframe with model input's data in airGR format from Create_Forcing_Inputs. (DatesR, P_1, P_2,...,P_n, E_1, E_2, ...E_n, Q). If Q is not available please provide only DatesR, P, and E.
Subbasins	Subbasins' shapefile. Must contain the following attributes: 'Area' (in km2), 'Region' (in letters), and 'COMID' (identifier number).
RunIni	Initial date of the model simulation in 'mm/yyyy' format.

RunEnd	Ending date of the model simulation in 'mm/yyyy' format.
WarmUp	Number of months for the warm-up period. NULL as default.
Parameters	Vector of initial model parameters (X1 and X2) and correction factors of P (fp) and E (fpe) in the following order: c(X1, X2, fp, fpe). In the case of existing more than one 'Region' (e.g. regions A and B) please provide model parameters in the following order: c(X1_A, X1_B, X2_A, X2_B, Fp_a, Fp_B, Fpe_A, Fpe_B).
Parameters.Min	Vector of minimum values of GR2M model parameters and correction factors in the following order: c(X1_min, X2_min, fp_min, fpe_min).
Parameters.Max	Vector of maximum values of GR2M model parameters and correction factors in the following order: c(X1_max, X2_max, fp_max, fpe_max).
Max.Functions	Maximum number of function evaluation for optimization. 5000 as default.
Optimization	Objective function for optimization (NSE, KGE, or RMSE).
No.Optim	Regions not to be optimized. NULL as default.

Value

List of optimal GR2M model parameters for each 'Region'.

Param: Best set of GR2M model parameters (sorted by 'Region').

Value: Final value of the objective function.

References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*. 2021; 13(8):1048. <https://doi.org/10.3390/w13081048>

Examples

```
# Optimize GR2M model parameters for a single 'Region' using the KGE metric
optim <- Optim_GR2MSemiDistr(Data=data,
                             Subbasins=roi,
                             RunIni='01/1981',
                             RunEnd='12/2002',
                             WarmUp=36,
                             Parameters=c(1000, 1, 1, 1),
                             Parameters.Min=c(1, 0.01, 0.8, 0.8),
                             Parameters.Max=c(2000, 2, 1.2, 1.2),
                             Max.Functions=1000,
                             Optimization='KGE')
best_param <- optim$Param
```

Routing_GR2MSemiDistr *Routing discharges for each subbasin.*

Description

Routing discharges for each subbasin.

Usage

```
Routing_GR2MSemiDistr(
    Model,
    Subbasins,
    Dem,
    AcumIni = NULL,
    AcumEnd = NULL,
    Save = FALSE,
    Update = FALSE
)
```

Arguments

Model	List of model results from Run_GR2MSemiDistr.
Subbasins	Subbasins' shapefile. Must contain the following attributes: 'Area' (in km2), 'Region' (in letters), and 'COMID' (identifier number).
Dem	Digital elevation model raster for the extent of the basin.
AcumIni	Initial date of the model routing in 'mm/yyyy' format. NULL as default
AcumEnd	Ending date of the model routing in 'mm/yyyy' format. NULL as default
Save	Boolean to save results as a text file in the 'Outputs' location. FALSE as default.
Update	Boolean for the updating mode where only the last month's values will be returned. FALSE as default.

Value

List of model routing outputs.

QR: Routed discharge timeseries for all subbasins in [m3/s].

Dates: Vector of dates of the simulation period.

COMID: Vector of identifier numbers for each subbasin.

Author(s)

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*. 2021; 13(8):1048. <https://doi.org/10.3390/w13081048>

Examples

[illegible]

Run_GR2MSemiDistr	<i>Run the GR2M model for multiple subbasins</i>
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Description

This function executes the GR2M monthly water balance model across multiple subbasins using a semi-distributed approach. It applies region-specific parameters and correction factors for precipitation and evapotranspiration, and optionally saves the results as text files.

Usage

```
Run_GR2MSemiDistr(
  Data,
  Subbasins,
  RunIni,
  RunEnd,
  Parameters,
  WarmUp = NULL,
  IniState = NULL,
  Save = FALSE,
  Update = FALSE
)
```

Arguments

Data	A data frame of model input data in the airGR format, as produced by Create_Forcing_Inputs. It must include columns: DatesR, P_1 to P_n, E_1 to E_n, and optionally Q.
Subbasins	A SpatVector object containing the geometries of subbasins. It must include attributes "COMID" (unique subbasin ID) and "Region" (region name/code).
RunIni	Simulation start date in the format "mm/yyyy".
RunEnd	Simulation end date in the format "mm/yyyy".
Parameters	A data frame containing GR2M model parameters and correction factors per region. Must have columns: Region, X1, X2, fp, fe.
WarmUp	Optional number of months to discard from the beginning of the simulation as warm-up. Default is NULL.
IniState	Optional list of initial states for each subbasin. Default is NULL.
Save	Logical. If TRUE, output time series will be saved as text files in the "Outputs/" folder.
Update	Logical. If TRUE and Save = TRUE, it will remove previous month's output files before saving the current ones.

Value

A list with the following elements:

PR Matrix of precipitation [mm/month] for each subbasin

AE Matrix of actual evapotranspiration [mm/month] for each subbasin

SM Matrix of soil moisture [mm/month] for each subbasin
RU Matrix of runoff [mm/month] for each subbasin
QS Matrix of discharge [m3/s] (not routed) for each subbasin
Dates Date vector for the simulation period
COMID Vector of subbasin COMIDs
EndState List of final state variables for each subbasin
SINK Optional. Data frame of simulated and observed outlet discharge, if observed Q is provided

References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. doi:10.3390/w13081048

Examples

```
# Run the GR2M model
model <- Run_GR2MSemiDistr(
  Data = data,
  Subbasins = roi,
  RunIni = "01/1981",
  RunEnd = "12/2016",
  Parameters = data.frame(
    Region = c("A"),
    X1 = 10.5,
    X2 = 0.8,
    fp = 1.1,
    fe = 1.0
  )
)
```

Index

Create_Forcing_Inputs, [1](#)

Optim_GR2MSemiDistr, [3](#)

Routing_GR2MSemiDistr, [4](#)

Run_GR2MSemiDistr, [6](#)